UIC Department of Chemical Engineering
UNIVERSITY OF ILLINOIS AT CHICAGO
COLLEGE OF ENGINEERING

ASSIGNMENT #01

Course: CHE 433: PROCESS SIMULATION WITH ASPEN PLUS
Term: Fall 2025
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Due Date: Friday, September 5th TA's Mailbox by 3:00PM

INSTRUCTIONS FOR ASSIGNMENT DELIVERY

- Prepare a typed report of your assignment.
- Include this cover page with your name(s) in the report.
- Upload your report as a PDF file by the Due Date and Time on the UIC Blackboard.
- Also place a hard copy of your report as a PDF file by the Due Date and Time in the TA's mailbox

- Assignments that are not typed will not be accepted.
- Late Assignments will not be accepted without prior approval.
- You may use the Internet as well as ChatGPT or other public tools as long as you place a written statement about their use in your assignment report

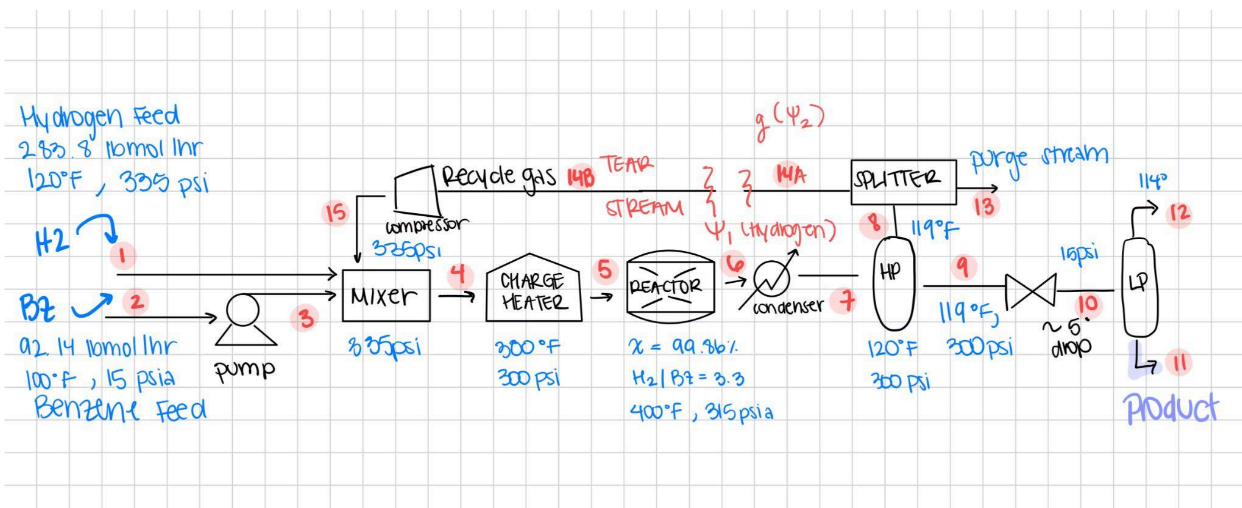
Problem A:

We are asked to provide a material balance for a cyclohexane production unit and a cost estimate.

1.

For our first tasks we must begin by adding the missing unit operations to our process flow diagram. It is evident to begin that our feed streams must be mixed together before entering the charge heater to ensure uniform composition. This will be achieved by adding a mixer. We also noticed that our streams do not begin at the same pressure either, with our benzene stream (BZ) having a low pressure. To increase the pressure of this stream we must add a pump and increase the pressure from *15 psia* to *335 psia*. This change in pressure increases our temperature very slightly. As we move along in the flow diagram, we must cool down our stream exiting the reactor using a condenser. This condenser is powered by cooling water and allows us to cool down our product stream significantly. This stream can now be fed into the HP separator. Additionally we must also add a pressure valve to lower and monitor the pressure of the stream leaving the high pressure separator drum so that it can enter the low pressure separator drum. Finally, we must add a compressor to raise the pressure of the recycle stream to 335 psia so it can enter the mixer.

AFTER ADDING MISSING UNIT OPERATIONS:



2.

For our next task we must perform a mass balance for our units. We do this in the ENG Calculations section of our CHE Calculator and we begin by numbering and defining our streams. We are given the flow rate of Benzene (BZ) and Hydrogen (H2) Feeds to be *92.14*

lbmol/h and *283.8 lbmol/h* respectively. We are told our H2 feed is a mixture with $z_{H2} = 0.975$, $z_{N2} = 0.005$, and $z_{CH4} = 0.020$.
Given *H2* flow rate and molar fraction, the flow rate of each component is calculated easily by the following equations:

$$F_{H2} = F_{H2}^{\circ} * z_{BZ} = 283.8\ lbmol/s * 0.975 = 276.705\ bmol/h$$

$$F_{N2} = F_{H2}^{\circ} * z_{N2} = 283.8\ lbmol/s * 0.005 = 1.419\ lbmol/h$$

$$F_{CH4} = F_{H2}^{\circ} * z_{CH4} = 283.8\ lbmol/s * 0.02 = 5.676\ lbmol/h$$

ENG Calculations	Peng-Robinson															
Molar Composition		Pump		Mixer	Charge heater	Reactor	Condenser	HP Separator		Valve	LP Separator			Splitter		Compressor
	Stream 1 (H2)	Stream 2 (BZ in)	Stream 3 (BZ out)	Stream 4	Stream 5	Stream 6	Stream 7	Stream 8 (bot)	Stream 9 (top)	Stream 10 (bot)	Stream 11 (bot)	Stream 12 (top)	Stream 13 (out)	Stream 14 A (tear)	Stream 14 B (recycle)	Stream 15
C6H6	0.0000	92.1400	92.1400	92.1455	92.1455	0.1290	0.1290	0.0056	0.1234	0.1234	0.1256	0.0009	0.0056	0.0055	0.0055	0.0055
H2	276.7050	0.0000	0.0000	304.0800	304.0800	28.0306	28.0306	27.9162	0.1145	0.1145	0.0021	0.1152	0.5412	27.3750	27.3750	27.3750
N2	1.4190	0.0000	0.0000	46.5114	46.5114	46.5114	46.5114	45.9839	0.5276	0.5276	0.0272	0.5135	45.9839	45.0924	45.0924	45.0924
CH4	5.6760	0.0000	0.0000	117.6861	117.6861	117.6861	117.6861	114.2244	3.4617	3.4617	0.4516	3.0964	114.2244	112.0101	112.0101	112.0101
C6H12	0.0000	0.0000	0.0000	3.6993	3.6993	92.0164	92.0164	3.7725	88.2440	88.2440	89.8381	0.6059	3.7725	3.6993	3.6993	3.6993
Zi	0.0000	1.0000	1.0000	0.1633	0.1633	0.0005	0.0005	0.0000	0.0013	0.0013	0.0014	0.0002	0.0000	0.0000	0.0000	0.0000
	0.9750	0.0000	0.0000	0.5390	0.5390	0.0986	0.0986	0.1455	0.0012	0.0012	0.0000	0.0266	0.0033	0.1455	0.1455	0.1455
	0.0050	0.0000	0.0000	0.0824	0.0824	0.1636	0.1636	0.2396	0.0057	0.0057	0.0003	0.1185	0.2795	0.2396	0.2396	0.2396
	0.0200	0.0000	0.0000	0.2086	0.2086	0.4138	0.4138	0.5952	0.0374	0.0374	0.0050	0.7148	0.6943	0.5952	0.5952	0.5952
	0.0000	0.0000	0.0000	0.0066	0.0066	0.3236	0.3236	0.0197	0.9543	0.9543	0.9933	0.1399	0.0229	0.0197	0.0197	0.0197
sum	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Stream Properties	Stream 1 (H2)	Stream 2 (BZ in)	Stream 3 (BZ out)	Stream 4	Stream 5	Stream 6	Stream 7	Stream 8 (bot)	Stream 9 (top)	Stream 10 (bot)	Stream 11 (bot)	Stream 12 (top)	Stream 13 (out)	Stream 14 A (tear)	Stream 14 B (recycle)	Stream 15
Molar Flow, lbmol/hr	283.8000	92.1400	92.1400	564.1223	564.1223	284.3736	284.3736	191.9025	92.4712	92.4712	90.4447	4.3319	164.5275	188.1823	188.1823	188.1823

A screenshot of our mass balance calculations is shown above

Pure benzene is added to our mixer along with a hydrogen mixture feed and a recycling stream that will be calculated later. The benzene feed must first go through a pump to ensure our feeds are equal in pressure. This does not change our flow rate, but increases the pressure from *15 psia* to *335 psia*. To find the flow rate exiting from our mixer we simply add *H2 (S1)*, *BZ (S2)*, and the recycle stream *(S15)*. We can assume *S15* is *0* for now and continue calculating, but account for it where it is necessary. We will use the newton method to calculate the recycle stream later on in the report.

The charge heater does not change our flow rate and we assume the above conditions apply as well. Flow rates of each component exiting the mixer are equal to the flow rates exiting the charge heater. We adjust our new temperature and pressure accordingly.

Charge heater:

- Temperature: *300*
- Pressure: *330 psia*

To calculate the flow rate of the components exiting the reactor *(S6)*, we must begin by calculating the extent of reaction. This indicates the amount of moles that have been transformed

by the reaction. Given the reaction Hydrogenation of Benzene:
$Benzene\ (BZ)\ +\ 3\ H_2 \rightarrow Cyclohexane\ (CH)$
The exit flow rate of benzene is determined by the following equation:

$$F_{BZ} = F_{BZ}^{\circ} - \varepsilon$$

Where F_{BZ}° is the amount of benzene entering the reactor, this includes the amount of benzene in the feed and in the recycling stream, ε is the extent of reaction given by multiplying the conversion by the stoichiometric coefficient. In this case, we are given the conversion of benzene in the reactor, X, to be 0.9986.

The exit flow rate of hydrogen is determined similarly by the following equation:

$$F_{H2} = F_{H2}^{\circ} - 3\ \varepsilon$$

We must add a coefficient of 3 to account for the stoichiometric coefficient. F_{H2}° is the amount of hydrogen entering the reactor and since hydrogen is present in the recycling stream, we must account for that as well as the initial amount of hydrogen in the feed stream.

The methane *(CH4)* and nitrogen *(N2)* present in the *H2* feeds are inert and therefore do not participate in the reaction. The amount of methane and nitrogen exiting the reactor is equal to that entering the feed plus the amount present in the recycling stream.

To calculate the amount of cyclohexane *(CH)* created by our reactor we multiply the amount of benzene entering the reactor by its conversion.

$$F_{CH} = F_{BZ}^{\circ} * X$$

Moving on to our next unit operation, the condenser. This unit is necessary to cool down our products and does not change our flow rates.

Next we must perform VLE calculations to determine the flow rates of each component in our product stream. This product stream is assumed to be a liquid-vapor mixture that must be separated through a high pressure flash drum first then a low pressure flash drum.

Given the following K values for the high pressure product separator (HP), we begin VLE calculations

- Benzene: *0.0217*
- Hydrogen*: 117.5*

- Nitrogen*: 42*
- Methane: *15.9*
- Cyclohexane: *0.0206*

The molar fraction of each component entering the HP separator can be calculated by the following equation:

$z_i = Fz_i / F_{tot}$

Where in this case Fz_i is the flow rate of component *i* and F_{tot} is the sum of all component flow rates *(S7)*

Next we calculate the x_i values for each component using the equation:

$$x_i = \frac{z_i}{1+(k_i-1)*\alpha}$$

We make an arbitrary guess for vapor fraction, α, and continue calculating the y_i values for each component

$y_i = x_i * K_i$

Once we have determined K_i, x_i, and y_i for each component we set up our Rachford Rice (RR) equation as follows:

$$f(\alpha) = \sum_{i=1}^{5} \frac{(K_i-1)*z_i}{1+(K_i-1)*\alpha}$$

Next, using goal seek we must find the vapor fraction that sets our RR equation equal to 0. By setting RR to equal 0 by changing α_1, we determine our vapor fraction to be 0.6748. Now we can calculate how much of the stream entering the HP separator is vapor & liquid.

- Vapor: *S7 ** α_1
- Liquid: *S7 - V*

Finally, we calculate the top and bottom exit streams of the HP separator by calculating $V* y_i$ and $L*x_i$ for each component. The top stream is assumed to be all vapor and the bottom stream is assumed to be all liquid.

The top stream of our HP separator will be fed to a separator which will purge and recycle accordingly. The bottom stream of our HP separator is fed to a valve which lowers and monitors pressure so the stream can be fed to a low pressure (LP) product separator. This valve does not change the composition of the HP bottom stream. Now to find the component flow rates exiting the LP separator similar VLE calculations as above must be done. The top stream (S12) is assumed to be vapor, and the bottom stream (S11) is our liquid product.

See excel sheet for VLE calculations below:

31	HP Drum K Values			C6H6	H2	N2	CH4	C6H12		
32	C6H6	0.0217	zi	0.0005	0.0986	0.1636	0.4138	0.3236	V	191.9025
33	H2	117.5000	ki	0.0217	117.5000	42.0000	15.9000	0.0206	L	92.4712
34	N2	42.0000	xi	0.0013	0.0012	0.0057	0.0374	0.9543		
35	CH4	15.9000	yi	0.000029	0.1455	0.2396	0.5952	0.0197		
36	C6H12	0.0206	yi*V	0.0056	27.9162	45.9839	114.2244	3.7725		
37			xi*L	0.1234	0.1145	0.5276	3.4617	88.2440		
38	alpha	0.6748								
39	RR	0.0000								
40	LP Drum K Values									
41	C6H6	0.3214	zi	0.0013	0.0012	0.0057	0.0374	0.9543	V	5.6132
42	H2	2461.0000	ki	0.3214	2461.0000	842.0000	306.0000	0.3010	L	86.8580
43	N2	842.0000	xi	0.0014	0.0000	0.0001	0.0019	0.9966		
44	CH4	306.0000	yi	0.000447	0.020268	0.092292	0.587024	0.299968		
45	C6H12	0.3010	yi*V	0.0025	0.1138	0.5181	3.2951	1.6838		
46			xi*L	0.1209	0.0007	0.0095	0.1666	86.5602		
47	alpha	0.0607								
48	RR	0.0000								

Coming back to the recycling stream we must finally calculate the stream we have been accounting for. We know that the top exit stream (S8) of our HP separator is sent to a splitter which will have 2 exit streams: a purge and a recycle stream. To aid in our calculations we have decided to split stream 14 into stream 14A and 14B. We first chose a recycle fraction for the splitter and formed 14B by multiplying this fraction by each component flow of S8. In parallel, we made initial guesses for the five component flowrates in stream 14A.

We then solved a set of eight equations and eight unknowns with Newton's method:

- Unknowns: the vapor fractions in the HP and LP drums, the recycle fraction, and the five component flows of stream 14A
- Equations: the five component residuals of stream 14A and 14B, the Rachford-Rice equations for the HP and LP drums, and the reactor H2:BZ error (3.3 - H2/BZ).

Newton's method simultaneously drives all residuals to zero, yielding the converged recycle fraction, flash vapor fractions, and the actual component flow rates in stream 14 (14A = 14B), as shown in the image below.

1	**ENG Calculations**	**Peng-Robinson**															
2	**Molar Composition**		**Pump**		**Mixer**	**Charge heater**	**Reactor**	**Condenser**	**HP Separator**		**Valve**	**LP Separator**			**Splitter**		**Compressor**
3		**Stream 1 (H2)**	**Stream 2 (BZ in)**	**Stream 3 (BZ out)**	**Stream 4**	**Stream 5**	**Stream 6**	**Stream 7**	**Stream 8 (top)**	**Stream 9 (bot)**	**Stream 10 (bot)**	**Stream 11 (bot)**	**Stream 12 (top)**	**Stream 13 (out)**	**Stream 14 A (tear)**	**Stream 14 B (recycle)**	**Stream 15**
4	C6H6	0.0000	92.1400	92.1400	92.1455	92.1455	0.1290	0.1290	0.0056	0.1234	0.1234	0.1209	0.0025	0.0001	0.0055	0.0055	0.0055
5	H2	276.7050	0.0000	0.0000	304.0800	304.0800	28.0306	28.0306	27.9162	0.1145	0.1145	0.0007	0.1138	0.5412	27.3750	27.3750	27.3750
6	N2	1.4190	0.0000	0.0000	46.5114	46.5114	46.5114	46.5114	45.9839	0.5276	0.5276	0.0095	0.5181	0.8914	45.0924	45.0924	45.0924
7	CH4	5.6760	0.0000	0.0000	117.6861	117.6861	117.6861	117.6861	114.2244	3.4617	3.4617	0.1666	3.2951	2.2143	112.0101	112.0101	112.0101
8	C6H12	0.0000	0.0000	0.0000	3.6993	3.6993	92.0164	92.0164	3.7725	88.2440	88.2440	86.5602	1.6838	0.0731	3.6993	3.6993	3.6993
9																	
10	Zi	0.0000	1.0000	1.0000	0.1633	0.1633	0.0005	0.0005	0.0000	0.0013	0.0013	0.0014	0.0004	0.0000	0.0000	0.0000	0.0000
11		0.9750	0.0000	0.0000	0.5390	0.5390	0.0986	0.0986	0.1455	0.0012	0.0012	0.0000	0.0203	0.1455	0.1455	0.1455	0.1455
12		0.0050	0.0000	0.0000	0.0824	0.0824	0.1636	0.1636	0.2396	0.0057	0.0057	0.0001	0.0923	0.2396	0.2396	0.2396	0.2396
13		0.0200	0.0000	0.0000	0.2086	0.2086	0.4138	0.4138	0.5952	0.0374	0.0374	0.0019	0.5870	0.5952	0.5952	0.5952	0.5952
14		0.0000	0.0000	0.0000	0.0066	0.0066	0.3236	0.3236	0.0197	0.9543	0.9543	0.9966	0.3000	0.0197	0.0197	0.0197	0.0197
15	sum	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
16	**Stream Properties**	**Stream 1 (H2)**	**Stream 2 (BZ in)**	**Stream 3 (BZ out)**	**Stream 4**	**Stream 5**	**Stream 6**	**Stream 7**	**Stream 8 (top)**	**Stream 9 (bot)**	**Stream 10 (bot)**	**Stream 11 (bot)**	**Stream 12 (top)**	**Stream 13 (out)**	**Stream 14 A (tear)**	**Stream 14 B (recycle)**	**Stream 15**
17	Molar Flow, lbmol/hr	283.8000	92.1400	92.1400	564.1223	564.1223	284.3736	284.3736	191.9025	92.4712	92.4712	86.8580	5.6132	3.7202	188.1823	188.1823	188.1823
18	Mass Flow, kg/s																
19	Volume Flow, m^3/s																
20	Temperature, F	120.0000	100.0000	100.0000		300.0000	300.0000	120.0000	119.0000	119.0000			114.0000				
21	Pressure, psia	335.0000	15.0000	335.0000	335.0000	330.0000	330.0000	330.0000		300.0000	15.0000						335.0000
22	Vapor Fraction																
23	Molecular Weight, kg/kmol																
24																	
25																	
26	**User Data and Calculations**																
27																	
28	BZ Conversion at Reactor	0.9986															
29	Extent of Reaction	92.0164															
30																	
31	HP Drum K Values				C6H6	H2	N2	CH4	C6H12						Recycle Residuals		
32	C6H6	0.0217		zi	0.0005	0.0986	0.1636	0.4138	0.3236		V	191.9025			C6H6	0.0000	
33	H2	117.5000		ki	0.0217	117.5000	42.0000	15.9000	0.0206		L	92.4712			H2	0.0000	
34	N2	42.0000		xi	0.0013	0.0012	0.0057	0.0374	0.9543						N2	0.0000	
35	CH4	15.9000		yi	0.000029	0.1455	0.2396	0.5952	0.0197						CH4	0.0000	
36	C6H12	0.0206		yi*V	0.0056	27.9162	45.9839	114.2244	3.7725						C6H12	0.0000	
37				xi*L	0.1234	0.1145	0.5276	3.4617	88.2440								
38	alpha	0.6748													Ratio error	0.0000	
39	RR	0.0000															
40	LP Drum K Values																
41	C6H6	0.3214		zi	0.0013	0.0012	0.0057	0.0374	0.9543		V	5.6132					
42	H2	2461.0000		ki	0.3214	2461.0000	842.0000	306.0000	0.3010		L	86.8580					
43	N2	842.0000		xi	0.0014	0.0000	0.0001	0.0019	0.9966								
44	CH4	306.0000		yi	0.000447	0.020268	0.092292	0.587024	0.299968								
45	C6H12	0.3010		yi*V	0.0025	0.1138	0.5181	3.2951	1.6838								
46				xi*L	0.1209	0.0007	0.0095	0.1666	86.5602								
47	alpha	0.0607															
48	RR	0.0000															
49																	
50	Splitter Recycle Fraction	0.9806															
51	H2 to BZ Ratio	3.3000															
52	Desired H2 to BZ Ratio	3.3000															

Finally, to calculate the purge stream, we subtract the component flow rates in S14 from the ones in S8.

We have now successfully completed component balances for each stream in our process flow diagram.

[3.]

To provide a preliminary cost of all utilities we are given the following base cost of utilities:

Cooling water: 0.354$/GJ

Natural Gas: 6.0$/GJ

Electricity: 0.06$/kWh

We assume our operating year is 8,000 hours and begin our calculations. Based on our flow diagram we know that we must burn natural gas to fuel our charge heater and we must find how much heat (q) is needed for operation. Similarly we must calculate how much cooling water is needed to operate the condenser. Finally our electricity costs will come after we determine the power supply needed for our process to operate.

To begin our calculations we calculated the molar enthalpies at the different unit operations in the next manner:

- Pump: We treated both pump streams as liquid benzene. After converting units, we determined the outlet temperature with PumpTemp() using the inlet state, outlet pressure, and pump efficiency. We evaluated the compressibility factor with ZComp() and the molar enthalpy with Enthalpy() at both inlet and outlet conditions, then multiplied by the benzene molar flow to get enthalpy rates. The difference across the pump gives the duty, which contributes to our electricity cost.
- Compressor: For the recycled gas, we estimated the outlet temperature with GasCompTemp() given the inlet state, outlet pressure, composition, and compressor efficiency. We calculated the compressibility factor with ZComp() at inlet and outlet, evaluated molar enthalpies with Enthalpy(), and formed enthalpy rates. Their difference is the compressor shaft power.
- Heater: We calculated the temperature into the heater by completing an enthalpy balance around the unit. We calculated the enthalpies of Stream 1, Stream 3, Stream 14, and Stream 4 by making a guess for the temperature. Then we use goalseek to solve for the temperature by setting the enthalpy balance to 0.
- Condenser: We calculated the inlet and outlet states similarly to the compressor and pump: consistent units, compressibility factor from ZComp(), molar enthalpy from Enthalpy(), then enthalpy rates. The outlet–inlet difference is the cooling duty.

See our calculations on excel below:

	ELECTRICITY						GAS		WATER	
Energy Balance Calculations	**pump in (S2)**	**pump out (S3)**	**valve in (S9)**	**valve out (S10)**	**compressor in (S14)**	**compressor out (S15)**	**heater in (S4)**	**heater out (S5)**	**condenser in (S6)**	**condenser out (S7)**
Temperature, C	37.7778	38.9641	48.3333	44.5268	48.8889	50.3295	45.7949	148.8889	148.8889	48.8889
Pressure, bar	1.0342	23.0974	22.7526	1.0342	22.7526	23.0974	23.0974	22.7526	22.7526	22.7526
Molar Flow, lbmol/hr	92.1400	92.1400	92.4712	92.4712	188.1823	188.1823	564.1223	564.1223	284.3736	284.3736
Molar Flow, kmol/s	0.0116	0.0116	0.0116	0.0116	0.0237	0.0237	0.0711	0.0711	0.0358	0.0358
Molar Enthalpy, kJ/kmol	51832.037	52110.370	-147580.819	-148286.134	-46440.0692	-46390.8616	-2478.545	2227.267	-63463.099	-70990.839
Enthalpy Flow, kJ/s	601.6382	604.8689	-1719.1949	-1727.4112	-1100.9323	-1099.7658	-176.1404	158.2831	-2273.5239	-2543.2003
Compressibility Factor	**0.0035**	**0.0786**	**0.0882**	**0.0041**	**0.9759**	**0.9760**	**0.9503**	**0.9823**	**0.9220**	**0.8036**
Duty [Power], kJ/s [kW]		3.2307		8.2163		1.1665		334.4235		269.6764

valve out calculations

	vapor	liquid
temperature	44.5268	44.5268
zcomp	0.9915	0.0041
enthalpy	-79833.0306	-151958.988
lever rule	-4846.0017	-142734.8173
balance	**0.0000**	

heater in calculations

	Stream 1	Stream 3	Stream 14	Stream 4
temperature	48.8889	38.9641	50.3295	45.7949
molar flow kJ/s	0.0358	0.0116	0.0237	0.0711
enthalpy (ig)	-814.3732	84165.9459	-46118.6862	-2047.0576
enthalpy flow	-29.1155	976.9527	-1093.3134	-145.4763
			balance	0.0000

Electricity, $/kW	0.0600		annual hours	8000.0000
Gas, $/GJ	6.0000			
Water, $/ GJ	0.3540			
		energy cons	annual cost	
Pump Power, kWh		25845.8407	$ 1,550.75	
Compressor Power, kWh		9332.3187	$ 559.94	
Heater Energy, GJ		9631.3967	$ 57,788.38	
Cooling Water, GJ		7766.6792	$ 2,749.40	
		total cost	$ 62,648.47	

4.

Finally, we are asked to propose a method to reduce utility costs. From the unit prices, we notice that natural gas is the most expensive utility in our process, so we propose an alternative to heat the feed going into the reactor. One that does not burn up the natural gas can save us up to $57,788. Alternatively, we can heat up our recycle stream prior to feeding it into our mixer. Using steam we can increase the temperature of the mixture which will allow us to burn less fuel. We can add heat to our streams in stages instead of relying on the charge heater to do so in one stage.